Synthetic local-first presentation pipeline

A hypothetical systems design for evidence-preserving technical talks

SYNTHETIC / HYPOTHETICAL TEST MATERIAL · 9-slide backend bake-off

Design the pipeline around evidence

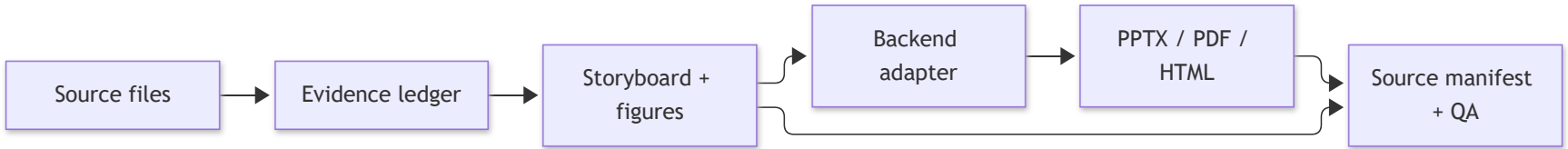
Problem

Presentation tools can draft and arrange slides, while scientific review needs exact data mapping, source lineage, and explicit uncertainty.

Test question

Can independent render backends produce the same editable talk without changing the evidence meaning?

Separate evidence from presentation layout



Evidence status and source IDs travel beside the rendered deck.

Keep figures deterministic

Figure field	Hypothetical input
Load tier A	120 req/s
Load tier B	240 req/s
Load tier C	360 req/s

These values are synthetic fixture values, not benchmark results. A renderer must preserve the values exactly and label the units.

Layout intent should survive backend changes

Assertion
One claim per slide

Evidence
Show source and data

Review
Keep caveats visible

SYNTHETIC DESIGN EXAMPLE · no measured results

Track limitations as structured output

- Unsupported chart type → explicit warning
- Raster-only element → object-level editability report
- Missing source ID → build error
- Unverified evidence → visible pending state
- Provider call → explicit opt-in and cost boundary

Compare capabilities, not marketing labels

Check	Required evidence
Export	PPTX produced by local command
Editability	Text, shapes, charts inspected in package
Fidelity	Every rendered page reviewed
Lineage	Source IDs preserved in sidecar
Operations	Version, dependencies, external services logged

A backend is useful only when its boundary is clear

Local renderer → repeatable exports without model calls

AI service → optional drafting, model/provider and cost disclosed

Scientific QA → stays outside the renderer and checks every output

SYNTHETIC / HYPOTHETICAL ARCHITECTURE

Recommendation for this test

Keep the evidence ledger and chart data in a backend-neutral contract.

Use mature renderers through isolated adapters; promote one only after same-input, editable-object and visual checks pass.

No backend comparison in this fixture establishes research performance.